Emergent Carrying Capacity, the Biocapacity Ratio, and the Productivity Illusion: Deficit-Driven Collapse in a Delayed Coupled Human–Environment Model

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Abstract

Coupled human–environment models typically treat carrying capacity as an imposed, static ceiling. We show that it is instead emergent from the interaction of environmental stock, technological productivity and per-capita demand; treating it as fixed can produce a false sense of environmental health. We develop a minimal deterministic model in which environmental assets (forests, soils, water systems) function as natural capital that yields a regenerative flow. Cumulative overshoot is tracked as ecological debt that erodes that yield; environmental regeneration and demographic response act through two distinct delays. Three results follow. First, under full harvest ($\sigma = 1$) the constant-parameter subsystem admits a one-parameter family of equilibria $P = B(A)/e$ rather than an isolated attractor; for the representative parameter set every interior point is monotonically unstable — a positive real eigenvalue for every delay, no imaginary-axis crossing, for $b/b_G + G'(A*) > r$. Collapse onset is therefore a structural vicious cycle, not a delay-ratio Hopf bifurcation, and is not controlled by a single stability index. Two mechanisms operate: debt-driven collapse even in the absence of delay, and a delay-amplified transient that shrinks the basin of attraction. Second, the operating boundary is the biocapacity ratio $R_B = 1$ (footprint equal to total biocapacity); the flow- yield ratio $R_A = 1$ is a leading but non-causal signal. Once the regeneration lag is long, neither ratio warns. Third, a sufficiently early technology wave can raise measured biocapacity while the stock continues to fall, but only inside a narrow window of about five years that exists solely for small initial deficits and vanishes beyond modest overshoot. This is a productivity illusion, not improving health: yield gains saturate while debt compounds, and the same technology simultaneously raises biocapacity, sustainable population and aggregate demand, tending to increase cumulative debt under overshoot. The decisive lever is the regeneration delay; neither a rising harvest nor an early- sounding ratio reliably signals collapse.

Keywords: * carrying capacity; ecological footprint; biocapacity; time delay; ecological debt; sustainability; delayed feedback

1 Introduction

Carrying capacity is one of ecology's most enduring yet elusive ideas. In classical population biology it is a fixed number — the ceiling that caps growth. Imported into global human ecology it is usually treated as a static boundary, the maximum number of people the Earth can support (Cohen, 1995). Such treatments overlook a feedback that matters: if a population overshoots its resource base, it degrades the very environment that sustains it and thereby lowers the future ceiling. The ceiling is a moving target. Imagine an orchard. Each year the trees bear fruit, and this fruit can be gathered without harming the trees themselves. This is a *flow*; ecologists call the annual regenerative flow of productive land and water biocapacity. The trees themselves remain standing — a stock of natural capital — and it is this stock that generates the flow. Taking the fruit at or below the rate at which the trees regrow uses the flow, which is biocapacity; cutting into the standing stock faster than it regenerates is liquidation of capital, which reduces the future flow rather than contributing to it. Suppose the harvesters take more than the orchard can regrow. At first nothing seems wrong — the trees still stand, the harvest still comes — but the surplus is not free. It is paid for by cutting branches, exhausting soil, and eventually felling trees. A harvest that looks stable is masking a shrinking asset. We call this the productivity illusion: the mistaken belief that a steady or rising harvest signals a healthy orchard. It sits at the heart of a long-standing debate in sustainability science between those who hold that technology can always substitute for natural capital and those who argue that substitution has hard limits. The Global Footprint Network turns this bookkeeping into two accounting quantities (Borucke et al., 2013). Biocapacity is the fruit; the ecological footprint is the demand placed on it. When the footprint exceeds biocapacity, an ecological deficit occurs, analogous to drawing down a bank account: the annual balance is a flow, but its cumulative effect erodes the underlying stock of natural capital, which in turn determines future biocapacity. The very quantity that defines the ceiling is eroded by sustained overshoot. Nor do the orchard and its harvesters respond instantly. Forest loss, soil degradation and climate change unfold over decades, while demographic patterns and social norms are slow to change. The system is therefore a delay differential equation, a class widely used to capture feedback delays in ecology (Kuang, 1993), and this is why civilisations can prosper for long periods while quietly undermining their productive base, only to suffer abrupt decline (Tainter, 1988; Diamond, 2005). The orchard also has an everyday analogue. A system can bear overload for a while — an elevator rated for ten people will still manage fourteen, showing little visible strain before the cable gives way — while its underlying integrity is quietly eroding. Judged by what it delivers (biocapacity, the harvest) the elevator still runs; judged by its reserve (the stock, the cable) it is already failing. This gap is the productivity illusion, and the question this paper addresses is not whether such masking can occur but when it can be sustained and when monitoring fails to reveal it. The wider literature frames the same problem. The most prominent global human–environment model is the World3 system of *The Limits to Growth* (Meadows et al., 1972, 2004) and *Dynamics of Growth in a Finite World* (Meadows et al., 1974), which produces overshoot scenarios through complex feedbacks among numerous interacting stocks; a rich literature on coupled human–environment and resource-harvest dynamics has since developed (Brander & Taylor, 1998; Anderies, 2000). Estimates of Earth's carrying capacity span an enormous range, from well under one billion to over one trillion people (Cohen, 1995; van den Bergh & Rietveld, 2004), and some scholars argue the very notion of a fixed carrying capacity is inapplicable to humans (Simon, 1996). Meyer (1994) and Meyer & Ausubel (1999) instead model carrying capacity as itself logistically varying, and Dasgupta (2021) formalises natural capital as a depreciating asset. These diverging perspectives all point to the same need: frameworks in which carrying capacity is not an exogenous input but an emergent property of the system. The model we develop makes the orchard precise. Biocapacity is the sum of a flow yield — separable from the stock, taken without reducing it — and the capital's growth — removable only by liquidating the base. The share of the two, the flow share, governs how the system responds to demand and tells us whether a rise in measured biocapacity is a true recovery or a mask over a shrinking stock. Because the stock is the only way to meet a shortfall, overshoot feeds damage back onto the supply: demand in excess of the flow yield is paid from the base, and cumulative overshoot accumulates as an ecological debt that steadily erodes yield. Two delays complete the picture — the regeneration delay before the orchard recovers, and the demographic delay before a population responds to a changing ceiling. Three conclusions follow, and each is a statement about the orchard. First, a sustainable balance is not one point but a family of balances, and once demand presses past the renewable flow the system begins a self-reinforcing loss that needs no oscillatory mechanism to start: the decline is structural, not a delicate, delay-tuned wobble. Second, the boundary that separates recovery from collapse is when total demand equals total biocapacity; the more commonly watched flow-only ratio is an early but non-decisive signal, and when the regeneration delay is long even that can stay silent. Third, an early enough technology wave can make measured biocapacity rise while the stock falls — but only within a narrow window, about five years, and only while the initial deficit is small; beyond a modest overshoot the window closes. The same technology that lifts biocapacity also lifts the sustainable population and hence total demand, so under overshoot it tends to increase cumulative debt. The message is therefore cautionary, and it names the levers precisely. The lever that matters is the regeneration delay — shorten it, and adopt the protective, effort-reducing institutional stance — while the levers that do not exist include rescue-by-stock and any early-warning signal. Rising biocapacity need not mean improving environmental health; it may be the productivity illusion that precedes a reckoning. The paper remains a conceptual, stylised illustration of the accounting logic, not a calibrated forecast: parameters are representative, the eight predictions it states are falsifiable hypotheses to be tested rather than results to be asserted, and the consequences of the model's deliberately minimal, Allee-free choices are discussed in §Model scope and §Limitations. —

2 Model formulation

2.1 Variables and units

Because A is in physical hectares it is independent of yield, so $b = B/A$ is definable and the decomposition $B = b \cdot A + b_G \cdot G(A)$ is identifiable. (With a stock measured directly in global hectares — as in the National Footprint Accounts — a hectare already embeds b, and the decomposition is not identifiable.) The full symbol table with the units of every quantity is given in the Supplementary Information (§S1.2).
Units — the flow/stock convention. E (footprint) and B (biocapacity) are per-annum flows and are written $gha \cdot yr^{-1}$; they are the amount of global-hectare-equivalent demand (or supply) in one year. The accumulated ecological debt D is their time-integral over the deficit, $D = \int [E - B]_+ dt$, and therefore carries $gha \cdot yr$ (global-hectare-years) — a *stock* of past overshoot, not a rate. The two are never interchanged: a $gha \cdot yr^{-1}$ rate enters the rate equations (4) and (7) directly, while the $gha \cdot yr$ accumulation D enters only the degradation law (8) as $e^{-\alpha D}$. The Global Footprint Network reports the footprint and biocapacity of a given year in global

Symbol	Meaning	Unit	State?
A	productive biocapacity area (the ”trees”)	ha	yes
P	human population	cap	yes
D	accumulated ecological debt (degradation channel)	gha·yr	yes
b	flow yield per unit area (fruit per tree)	gha·ha^{-1}·yr^{-1}	derived
b_G	value of one hectare of standing stock	gha·ha^{-1}	parameter
$G(A)$	regeneration / recruitment rate	ha·yr^{-1}	no
B	biocapacity = $bA + b_G G(A)$	gha·yr^{-1}	no
E	footprint = eP	gha·yr^{-1}	no
K	carrying capacity = B/e	cap	algebraic, not a state
e	per-capita footprint	gha·cap^{-1}·yr^{-1}	param
ρ	regeneration rate	yr^{-1}	param
r	per-capita population growth rate	yr^{-1}	param
α	degradation rate	(gha·yr)$^{-1}$	param
η	debt-repayment rate	yr^{-1}	param
τ_g	recruitment (regeneration) delay	yr	param
τ_p	demographic (generation) delay	yr	param
A_{max}, b_0, A_{ext}	carrying-area ceiling, base yield, extinction floor	ha, gha·ha^{-1}·yr^{-1}, ha	param
T_b	technology-driven yield step (the yield-enhancing wave $T_b(t)$)	gha·ha^{-1}·yr^{-1}	param
Δb	amplitude of the yield wave (asymptotic rise of T_b)	gha·ha^{-1}·yr^{-1}	param
κ	steepness / rate of the yield wave	yr^{-1}	param
t_{wave}	time of the yield wave’s midpoint	yr	param

hectares (gha) with the "per year" basis left implicit; the model makes that basis explicit ($gha \cdot yr^{-1}$) so that E and B can be differentiated in time. The per-capita footprint e is likewise $gha \cdot cap^{-1} \cdot yr^{-1}$ (per-person annual demand), so $E = eP$ is a flow.

2.2 Governing equations

```
Regeneration:  G(A) = ρ A (1 − A/A_max)                         (1)
Biocapacity:   B    = b A + b_G G(A)                           (2)
Footprint:     E    = e P                                      (3)
Stock:         dA/dt = G(A(t−τ_g)) − [E(t) − σ b A(t)]₊ / b_G   (4)
Population:    dP/dt = r P(t) [1 − P(t) / K(t−τ_p)]             (5)
Carrying cap:  K    = B / e                                    (6)   [algebraic]
Debt:          dD/dt = [E − B]₊ − η D                          (7)
Degradation:   b    = (b₀ + T_b(t)) e^{−α D(t)}                (8)
Technology:    T_b  = Δb / (1 + e^{−κ(t−t_wave)})              (9)
```

Notation and conventions. Four conventions are used throughout and are stated here so that the equations are unambiguous:

- σ **is a dimensionless fraction**, $0 \leq \sigma \leq 1$, the human-available share of the flow. $\sigma = 1$ means the *entire* flow yield bA may be harvested without damaging the base; $\sigma < 1$ reserves a share $(1-\sigma)B$ (a Half-Earth / reservation policy, §9), and σ is then set together with the population cap so that $E \leq \sigma \cdot B$. The factor $\sigma b A(t)$ in (4) is the safe flow harvest; only the excess $[E(t) - \sigma b A(t)]_+$ draws down the stock. **Consequence:** the one-parameter equilibrium family $P = B(A)/e$ of §4.3 and §4.5, and the neutral continuum $D(0) = 0$ on which it rests, are derived under $\sigma = 1$. Under a reservation $\sigma < 1$ the equilibrium locus shifts to $P = (\sigma b A + b_G G(A))/e < B/e$, i.e. reserving a share removes $(1-\sigma)bA$ from the sustainable population (verified in §4.3 / §S3.3).
- $[x]_+$ **is the positive part,** $[x]_+ = max(x, 0)$**.** In (4) and (7) it enforces the *one-sided* deficit mechanism: only a positive shortfall above the harvestable/share-adjusted flow liquidates stock or accrues debt; a surplus ($E \leq \sigma b A$, $E \leq B$) contributes nothing.
- b_G **is a conversion factor between the two "books", not a yield.** It is the value (in global hectares) of one hectare of standing stock, with units $gha \cdot ha^{-1}$. In (2) the capital-growth term $b_G G(A)$ has units $gha \cdot ha^{-1} \cdot ha \cdot yr^{-1} = gha \cdot yr^{-1}$, matching the flow term bA (b in $gha \cdot ha^{-1} \cdot yr^{-1} \times A$ in ha), so B is a flow and the two terms are commensurable. In (4) dividing the overshoot $[E - \sigma b A]_+$ (a $gha \cdot yr^{-1}$ flow) by b_G ($gha \cdot ha^{-1}$) yields a **stock-loss rate** in $ha \cdot yr^{-1}$, so `1/b_G` is the harvest coefficient $\gamma = 1/b_G = 1/V$ of §4.1 (V = standing biomass $\div$ annual production, $\approx$ 20–100 yr). b_G is therefore **not** dimensionless; it is the book-conversion rate that turns a flow deficit into capital loss.
- **What debt degrades.** Equation (8) degrades **only the flow-yield coefficient** b (via $e^{-\alpha D}$): cumulative ecological debt erodes the surviving stock's *productivity per unit area*, not the stock itself. It does **not** degrade b_G (the standing-stock value), ρ (the regeneration rate), or A_{max} (the carrying-area ceiling); those remain fixed at their baseline values. This is Assumption (8) and is a deliberate scope: the model keeps the degradation channel (flow productivity) separate from the regeneration channel (rate and ceiling), so that the two do not double-count. A debt dependence on $G(A)$ or b_G would be an additional, separately-argued extension.

Delay structure. The two delays enter **only** where stated. Regeneration is delayed, $G(A(t-\tau_g))$ in (4): the environment responds to its own state τ_g years earlier. Population response is delayed, $K(t-\tau_p)$ in (5): the population adjusts to the carrying capacity it "knew" τ_p years ago. Debt formation (7) and harvest (via $E = eP$ in (4)) act on **current** values — the lags are in recruitment and demography, not in the depletion or debt terms. Crucially, $K = B/e$ (6) is **algebraic and not itself delayed**: carrying capacity is computed from the *current* state $B = bA + b_G G(A)$, and the delay enters only through the population's response to a delayed value of it, $K(t-\tau_p)$. The system is genuinely three-dimensional in the states A, P, D; K carries no dynamics of its own. Equation (4) is the central structural assumption. Demand is met first from the flow yield bA (the fruit — leave the tree); only the shortfall $[E - bA]_+$ liquidates the standing stock, at rate `1/b_G`. This is the deficit mechanism, and it is why biocapacity appears in the stock equation: the two flow accounts are no longer parallel and unlinked. In the pure-capital limit ($\psi \to 0$, equivalently the saw-log form) equation (4) reduces to the standard gross-depletion harvest equation with harvest coefficient $\gamma = 1/b_G$. The model is well posed at the boundary. The original `dP/dt` diverges ($P/K \to \infty$ as $K \to 0$); we impose an extinction floor $A_{ext} > 0$ so that $K \geq b \cdot A_{ext}/e > 0$ is bounded and clamp $A \geq A_{ext}$, $P \geq 0$. "Collapse" is then a well-defined approach to the extinction boundary, not a numerically clamped blow-up (see §Numerics). The three state variables A, P, D live in three distinct "books", and the model is written so these are never mixed: A is the capital book (the standing base, removed only by liquidation); the flow yield bA is the flow-yield book (the fruit, separable from the base); and D is the degradation/debt book (the erosion of the surviving stock's yield). Conservation is a property of the incidence structure (each outflow is assigned to exactly one donor book) and positivity follows from donor limitation (an outflow vanishes when its donor compartment is empty). Any parameter pinned at an optimisation bound is reported as a declared fit defect rather than silently as a calibration. —

3 Assumptions

Each equation carries an explicit modelling choice.

- **(1) Logistic regeneration** — symmetric self-limiting growth; a modelling convenience, not a law. The point `A_max/2` is the maximal-growth point, not a threshold.
- **(2) Biocapacity is additive flow + capital growth** — the flow is separable (crops, orchard); the capital growth is removable only by drawing down the stock (forest, fish). The composition is captured by the flow share $\psi = bA*/B*$, which is the master parameter separating the two ratios R_A and R_B (§4.5) and locating the masking illusion (prediction 5).
- **(3) Per-capita footprint is constant** — e (and the per-capita requirement it represents) is held constant as a baseline modelling choice, since this is the only way to isolate the stock–flow–demand feedbacks; endogenising e as a time-varying per-capita requirement is an offered extension. The model does not carry a co-evolving per-capita requirement, and this is stated explicitly rather than left implicit.
- **(4) Deficit-driven, immediate depletion** — liquidation is immediate; the delayed response is in recruitment τ_g (a tree takes $\tau_g \approx 20\text{–}80 yr$ to bear fruit), not in the depletion term.

- **(5) Demographic delay on carrying capacity** — the population responds to the delayed carrying capacity $K(t-\tau_p)$ (the conditions a cohort "knew"), Hutchinson-style.
- **(6) K is algebraic, not a state** — a function of the state; the system is genuinely 3-D (A, P, D).
- **(7) Debt accrues only in overshoot, repayed at η — η** is primary (it decides whether an equilibrium exists). We set $\eta = 0.05yr^{-1}$, a minimal environmental regeneration / degradation-removal rate (slow, ≈ 20-yr timescale); the model is insensitive to η over a modest range provided $\eta > 0$.
- **(8) Degradation erodes the surviving stock's yield** — a separately-evidenced channel (soil fertility, over-picking), not double-counting the trees removed by A.
- **(9) Technology is bounded** — a logistic wave; saturation is why the masking is transient.

Delay asymmetry. Depletion/liquidation is immediate while functional degradation is a separate, slower state. We justify this asymmetry explicitly and note that a third lag τ_D could be added ($dD/dt = [E(t-\tau_D) - B]_+ - \eta D$) to keep the delay structure fully consistent. We also declare the aggregation of the governance clock: τ_p lumps several institutional time objects (observation, assessment, review, decision/deployment, ecological response, memory) and is not a single measured lag. —

4 Analytic results

This section develops the analytic core in two registers that read the same object differently. The dynamical view (§4.1–§4.4, §Numerics, §Discussion) asks how the system behaves; the observational view (§4.5, §Predictions, §Discussion) asks what a policy-maker can watch. The thesis that holds them together is stated in §Presentation and in the abstract: the balance point is $R_B = 1$; $R_A = 1$ is a leading but non-causal signal; and when the regeneration lag is too long neither ratio warns — the lag, not the ratio, is the controlling variable.

4.1 Carrying capacity, MSY, and the emergent ceiling

$K = B/e$ is emergent (it moves with A and b). Because the capital growth $b_G G(A)$ is positive only for $0 < A < A_{max}$, the total sustainable-yield curve $B(A) = bA + b_G G(A)$ has an interior maximum — the maximum sustainable yield (MSY): via $dB/dA = b + b_G \rho(1 - 2A/A_{max}) = 0$,

```
A*  = A_max (b + b_G ρ) / (2 b_G ρ)
B_max = A_max (b + b_G ρ)² / (4 b_G ρ)
```

(this derivation is set out in full in §S3.1). So the sustainable equilibrium is interior ($A* < A_{max}$), not A_{max} — a steady harvest of the capital growth keeps the stock below its unmanaged maximum. This is the surplus-production / MSY structure of Schaefer (1954), the same symmetric intrinsic-growth curve that gives $MSY = rK/4$. Only in the flow-only limit ($b_G G \ll bA$, the orchard) does the stock tend to A_{max} with a zero harvest drain. **The interior-MSY statement is regime-conditional.** $A* < A_{max}$ holds only in the capital-dominated regime, i.e. when $b_G\rho > b$ (equivalently $\psi \to 0$). When $b_G\rho < b$ (the flow-dominated regime, $\psi \to 1$) $B(A)$ is monotone increasing on $[0, A_{max}]$ and there is no interior maximum: the sustainable point is the boundary A_{max}. This is precisely the regime the baseline parameter set sits in ($b_G\rho = 0.8 \cdot 0.05 = 0.04 < b = 0.5$), so the baseline sustainable state is the boundary A_{max}. The regime must be stated before quoting $A*$/B_{max}; they are the capital-only limit, not universal. **The flow share** $\psi = bA*/B*$ **interpolates between the two limit cases.** Report its regime dependence explicitly, as it governs which analytic core applies:

- $\psi \to 0$ (capital-dominated): interior $A*$; the stability classification of §4.3 applies.
- $\psi \to 1$ (flow-dominated, the orchard limit): A_{max} at sustainability, a liquidation mask, and the vicious cycle — the $a_{11} = \rho(1 - 2A*/A_{max}) + b/b_G$ liquidation feedback.
- $0 < \psi < 1$: interior $A*$ and liquidation feedback coexist. This is the general object, and the increasing-harvest and orchard models are recovered as limits.
- $\gamma = 1/b_G = 1/V$, the orchard's salvage value (V = standing biomass ÷ annual production, $\approx$ 20–100 yr). The capital-harvest coefficient is therefore measurable and is not a free parameter.
- ψ is a testable prediction (prediction 5). Small ψ (flow yield is a small share, cropland/grazing) puts the model near the capital limit, where a smooth decline is stabilised by demographic feedback and the illusion is small. Large ψ (forest/fishery) makes the overshoot invisible until demand exceeds the flow yield, then liquidation with a threshold $A_c(E)$. The model therefore predicts the masking illusion is more visible in flow-dominated (high-ψ) systems than in capital-dominated (low-ψ) ones.

Regime $\to$ outcome. The single testable condition $b_G\rho \lesseqgtr b$ (equivalently $\psi \lesseqgtr 1/2$) decides everything downstream:

A parameter sitting exactly on the $b_G\rho = b$ boundary is a non-generic choice and should be stated and perturbed off.

4.2 Fixed-liability threshold and the saddle-node = MSY

For a fixed liability E, the deficit-region equation $dA/dt = 0$ is a quadratic in A with an unstable root (the separatrix), and the saddle-node is exactly the MSY:

```
A_c(E) = [ (b_G ρ + b) − √((b_G ρ + b)² − 4 b_G ρ E/A_max) ] A_max / (2 b_G ρ)
E_sn   = B_max = A_max (b + b_G ρ)² / (4 b_G ρ)          (safe basin vanishes at MSY)
```

(the fixed-liability threshold is derived in §S3.2). A fixed liability above B_{max} cannot be sustained by harvesting regrowth. The threshold is emergent and E-dependent (no Allee term needed), located at $A_c(E)$, never at `A_max/2`. When realised, the fold is a saddle-node (fold) bifurcation — the safe basin vanishes as E crosses B_{max}, the catastrophic-shift / critical-transition structure of alternative-stable-state ecology (Scheffer, Carpenter, Foley, Folke & Walker 2001): a loss of resilience precedes the switch, and the switch is abrupt and hard to reverse, not

Regime	Condition	$B(A)$ shape	Sustainable point	MSY	Collapse/mask behaviour
capital-dominated	$b_G\rho > b$ ($\psi \to 0$)	interior maximum	interior $A* < A_{max}$	interior MSY $A*$, B_{max}	smooth decline stabilised by demographic feedback — illusion small
flow-dominated (baseline)	$b_G\rho < b$ ($\psi \to 1$)	monotone ↑	boundary A_{max}	none interior (max at A_{max})	liquidation with threshold $A_c(E)$; overshoot invisible until demand exceeds flow yield — illusion more visible
marginal	$b_G\rho = b$	max at the boundary	$A* = A_{max}$ (interior max merges with boundary)	boundary A_{max}	$\psi \to 1$

because of an imposed Allee minimum but because the stable and unstable equilibria first meet at the fold and then destroy each other. **The fold is regime-scoped.** It is realised only in the capital-dominated regime ($b_G\rho > b$), where $B(A)$ has an interior maximum. There $dB/dA = 0$ exactly at the interior MSY, and for E just below B_{max} there are two fixed points — one unstable ($B'(A) > 0$) and one stable ($B'(A) < 0$) — that merge at the saddle-node (verified numerically: $B'(A*) = 0$; two roots with opposite-sign B' for $E < B_{max}$). At baseline ($b_G\rho = 0.04 < b = 0.5$), $B(A)$ is monotone increasing on $[0, A_{max}]$ with $B'(A) > 0$ everywhere, so no interior fold is realised: the fixed-liability equilibria are single-valued and unstable, and the stable object is the boundary A_{max}. **The fold must not be conflated with the τ_g cliff.** As we show in §Numerics and §Discussion, the τ_g transition is a basin-boundary crisis, not a fold: the leading eigenvalue is constant and positive ($Re\lambda \approx +0.62$) for every τ_g, so no eigenvalue crosses zero as $\tau_g \to 20$ and there is no critical-slowing-down / early-warning precursor (the P-relaxation return time is flat ≈ 150 yr for $\tau_g = 0$–18, then the attractor vanishes abruptly rather than slowing). Scheffer's "loss of resilience precedes the switch" therefore describes only a capital-dominated E-fold; it is the wrong description of the delay transition. The two must be told apart (§Discussion).

4.3 Stability; the vicious cycle; the stability classification

The deficit = stock-decline identity. When demand exceeds the flow yield ($E > bA$), Eq. (4) reduces exactly:

```
dA/dt = G(A(t−τ_g)) − [E − bA]₊/b_G  =  (B̃ − E)/b_G
```

because $[E - bA]_+ = E - bA$ there, and the lag-adjusted biocapacity is $\tilde{B} = bA + b_G G(A(t - \tau_g))$ (so $\tilde{B} = B$ only when regeneration is evaluated at the delayed state). This is the algebraic identity that makes "deficit-driven" rigorous: the shortfall between biocapacity and footprint, not the gross harvest, drives stock decline, and the model's stock build-up exactly cancels against its flow term. Corollary (in the R_B form of §4.5): $dA/dt < 0 \iff E > \tilde{B}$, and at equilibrium $E = \tilde{B}$, so $R_B = E/\tilde{B} = 1$ at the balance point. The $R_B = 1$ locus of §4.5 is exactly the neutral continuum on which the characteristic equation has $D(0) = 0$ (§Numerics) — the separator and the continuum are one and the same object, seen from the ratio (monitoring) and spectral (dynamical) views. Linearising (4)+(5) in the deficit region gives (at the interior point, under σ = 1 so that the equilibrium family $P = B(A)/e$ is well defined)

```
a₁₁ = ρ(1 − 2A*/A_max) + b/b_G      ;  a₁₂ = −e/b_G      ;  a₂₂ = −r
det = r·ρ·A*/A_max > 0  (never a saddle)   →   zero-delay condition is  a₁₁ < r  (NOT a
```

Sufficient condition for the monotone instability (the "positive real eigenvalue for every delay" claim). Linearising gives the two-delay characteristic equation $D(s; \tau_g, \tau_p) = (s - a_1 e^{-s\tau_g} - a_3)(s + r) - a_E a_4 e^{-s\tau_p} = 0$ with $a_1 = G'(A*)$, $a_3 = b/b_G$, $a_E = -e/b_G$, $a_4 = rK'(A*)$, and the equilibrium condition $P* = K(A*)$ forces $-a_E a_4 = r \cdot S$, where $S = a_1 + a_3 = G'(A*) + b/b_G$. Then:

- $D(0) = 0$ on the whole family — the neutral continuum, so there is **no isolated interior attractor**.
- With no delay the non-neutral root is $\lambda = S - r$. Hence a **positive real root** exists at zero delay exactly when $S > r$, i.e. $G'(A*) + b/b_G > r$.
- The root persists **for every delay** provided $F'(0) = r - S + ra_1\tau_g - rS\tau_p < 0$. At baseline ($a_1 = -0.0167$, $S = 0.6083$, $r = 0.02$) this holds over the whole $(\tau_g, \tau_p) \in [0, 400]^2$ plane (max $F'(0) = -0.5883$), so a positive real eigenvalue is guaranteed for every delay combination (verified: leading $Re\lambda = +0.625$ real across the plane; the exact $s = i\omega$ crossing-curve scan finds zero imaginary-axis crossings).

So the claim is conditional, not parameter-free. A positive real eigenvalue for every delay is guaranteed only when $S = b/b_G + G'(A*) > r$ (and the delay condition above); it is *not* a universal theorem for arbitrary parameter values. It is stated here for the representative parameter set, as checked numerically in §Numerics and §S5. Under a reservation $\sigma < 1$ the equilibrium locus is no longer $P = B(A)/e$ but $P = (\sigma bA + b_G G(A))/e < B/e$, and the whole family (and hence the neutral continuum) is shifted — the $\sigma = 1$ restriction is required for the results of §4.3 and §4.5 as stated (verified numerically: at $\sigma = 0.9$, $dA/dt(P = B/e)$ is negative for every interior $A*$).

- **The vicious cycle is real and quantitative.** a_{11} gains the `+b/b_G` term. For liabilities that scale with the stock ($E = f \cdot bA$) the interior point is $A* = A_{max}[1 - (f - 1)b/(b_G\rho)]$, and it is self-sustaining (the environment's own mode goes locally runaway, $a_{11} > 0$) precisely when $(2f - 1)\nu > 1$, where $\nu = b/(b_G\rho)$. With b_G the standing-stock value (20–100 yr) and realistic $\rho \approx 0.02$–$0.1 yr^{-1}$, ν is $O(0.1$–$2)$; treat $(2f - 1)\nu > 1$ only as an estimated indicator, not a measured threshold, because f and the liability elasticity are set by the model. The "vicious cycle" therefore describes Eq. (4) and not a gross-depletion form.

- **Stability classification (sign-corrected).** For $\rho \gg r$, the 2-D two-delay system reduces to a scalar two-gain delayed logistic with control $\chi = q/(\rho - 2q)$, $q = \gamma e b_0/r_{opt}$ (depletion pressure): $\chi > 1$ ($\Lambda > 0$) $\Rightarrow$ τ_g-only Hopf: $\omega = r\sqrt{}(\chi^2 - 1)$, $\tau_g* = arccos(-1/\chi)/\omega$. $\chi < 1$ ($\Lambda < 0$) $\Rightarrow$ τ_p-only Hopf: $\omega = r\sqrt{}(1 - \chi^2)$, $\tau_p* = arccos(-\chi)/\omega$. $\chi = 1$ ($\Lambda = 0$, i.e. $\rho = 3q$) $\Rightarrow$ neither; two-delay boundary $s = \pi/\omega$, $\omega = 2r \cdot cos(\omega d/2)$ $\Rightarrow$ $s \approx \pi/(2r) = 78.5yr$ at $d \to 0$. The sign structure is exact (from the $|Q| = |P|$ elimination); only $\omega*, \tau*$ are $O(r/\rho)$ approximate, and the fast–slow reduction is valid only while $a_{11} \leq 0$ (else use the full 2-D transcendental equation, where the A-mode can itself oscillate). On the constant-parameter S0 of the present model this Hopf classification is not realised: the exact $[\cdot]_+$ gives a monotone positive-real eigenvalue with no imaginary-axis crossing for every delay (§Numerics). The classification above describes the fast–slow reduction and does not predict the actual (structural) instability. **Baseline sits at a knife-edge.** Baseline $\chi = 1$ because $\rho = 3q$, i.e. $\Lambda = 0$, the measure-zero surface where the two gain surfaces $r^2 a_{11}^2 = (\gamma e a_{21})^2$ are equal ($g_M = g_P$). This is a non-generic, unexplained parameter choice, and it is precisely why "no single delay destabilises" holds at baseline. We do not justify it; we report instead the generic result that follows from the sign of Λ: perturbing off the knife-edge gives exactly one lag destabilising — τ_g-only if $\chi > 1$ ($\Lambda > 0$), τ_p-only if $\chi < 1$ ($\Lambda < 0$). Any scan range and any "no single delay destabilises" claim must be stated as a function of Λ/χ, never as a parameter-free statement (§Numerics).

4.4 The complete dimensionless group set

$\chi = q/(\rho - 2q)$ is the cleanest single control (it matches the scalar two-gain picture), but the paper carries the complete non-dimensionalization as the full generality statement: $\hat{t} = rt$, $a = A/A_{max}$, $p = P \cdot r_{opt}/(b_0 A_{max})$, and the groups $s = \rho/r$, $g = \gamma b_0 f/\rho$, $f = e/r_{opt}$, θ, plus scaled delays $\hat{\tau}_M = r\tau_g$, $\hat{\tau}_P = r\tau_p$ (full detail and scaling in §S2). We present both: χ for the clean stability-sign rule, and the six-group set for full generality — they are complementary, not competing.

4.5 Macro-ratio safe operating space (R_B vs R_A)

In addition to the dimensionless groups, the bookkeeping can be reported as two observable ratios that summarise the safe-operating picture. Define the biocapacity ratio and the flow-yield ratio:

```
R_B = E / B           (footprint ÷ total biocapacity)
R_A = E / (b A)       (footprint ÷ flow yield)
ψ   = bA / B          (flow share, 0 ≤ ψ ≤ 1)
R_A = R_B / ψ
```

The operating boundary is $R_B = 1$, and it is exactly the neutral equilibrium family of §4.3 $P = B(A)/e$. From the deficit identity $dA/dt = (\tilde{B} - E)/b_G$ (§4.3), $dA/dt < 0 \iff E > \tilde{B}$, so the stock declines iff footprint exceeds total biocapacity. At equilibrium $E = \tilde{B}$ so $R_B = 1$; this is simultaneously the bookkeeping trigger, the macro-ratio safe-operating boundary, and (as the family $P = B(A)/e$) the basin separator. Verified onset (baseline S0, documented A_0 grid): the recover/collapse boundary is crossed at $R_B = 1.000$ for every $A_0 \in 0.30, \ldots, 1.05$, while the flow-yield ratio there is $R_A = 1.01$–1.06. $R_A = 1$ **is necessary but not sufficient.** $R_A = 1$ means $E = bA$, i.e. demand equals the flow only. A system sitting between $R_A = 1$

and $R_B = 1$ still regenerates ($dA/dt > 0$), because the capital growth $b_G G(A)$ is untouched. $R_A > 1$ is therefore not the collapse trigger, and the two ratios are not interchangeable: R_A is a leading, non-causal signal; R_B is the trigger. **The gap between them is the flow share, which is the master parameter.** $R_A = R_B/\psi$, so the two ratios coincide in the flow-dominated limit ($\psi \to 1$, baseline) and separate as $\psi \to 0$ (capital-dominated). Regime-scoped closed form (at a sustainable interior MSY): $\psi* = 2/(1 + b_G\rho/b)$, $R^e_A q = (1 + b_G\rho/b)/2$. Verified across regimes: $b_G\rho/b = 0.05$ (flow-dominated, no interior MSY, $\psi \to 1$, $R_A = 1$); $2.0 \to \psi* = 0.667, R^e_A q = 1.50$; $37.5 \to \psi* = 0.052, R^e_A q = 19.25$; $150 \to \psi* = 0.013, R^e_A q = 75.5$. The single regime index $b_G\rho/b$ therefore governs the gap between R_A and R_B ($\psi* = 1/R^e_A q = 2/(1 + b_G\rho/b)$). **A sustainable system can run at $R_A > 1$.** In the capital-dominated regime the balance point is $R_B = 1$ with $R_A > 1$ (footprint > flow yield, the buffer intact). $R_A > 1$ is not a warning; the balance point is $R_B = 1$ in every regime. **Temporal-lead nuance.** Because $R_A = R_B/\psi$, R_A crosses 1 earlier than R_B by the factor $1/\psi$, but its lead is ≈ 0 in flow-dominated systems and grows only with capital dominance. A large lead is therefore a signature of the capital-dominated regime, not a general warning property. **Fold $\leftrightarrow$ $R_B = 1$ (regime-scoped).** The §4.2 fixed-liability fold threshold $E_{sn} = B_{max} = A_{max}(b + b_G\rho)^2/(4b_G\rho)$ is precisely the $R_B = 1$ boundary evaluated at $(A*, E = B_{max})$. Verified (capital-dominated $b = 0.02$): interior MSY $A* = A_{max}(b + b_G\rho)/(2b_G\rho) = 0.900$, $B(A*) = B_{max} = 0.0270$, $E_{sn} = 0.0270$, so $R_B = E_{sn}/B_{max} = 1.000$ exactly — the fold and the balance point are the same object. This fold is realised only when $b_G\rho > b$, the identical scoping as the §4.1 regime table and §4.2; at baseline it is absent. **The separator degrades with the lag — two sides of one coin.** At short/no delay the neutral family $P = B(A)/e$ (i.e. $R_B = 1$) separates recover from collapse at balanced accuracy 99.2 % (no-delay) $\to$ 81.2 % ($\tau_g = 30$); a linear functional is chance at $\tau_g = 30$ (balanced 50.0 %, raw 94.7 % = the majority-collapse class, a class-imbalance artefact). So $R_B = 1$ is necessary for recovery but not sufficient once the lag is too long — the operating boundary (here) and the basin crisis (§Numerics, §Discussion) are complementary statements about one object. The family $P = B(A)/e$ is the separator, not a "first integral". **One object, many faces.** The single scalar $R_B = 1$ is simultaneously the bookkeeping trigger ($dA/dt < 0 \iff E > \tilde{B}$, §2.2, §4.3); the basin separator $P = B(A)/e$ (§4.3, §Numerics — the neutral continuum); the interior-MSY fold threshold of §4.2, only when $b_G\rho > b$ (absent at baseline); the target of prediction 8 (§Predictions); and the object whose long-τ_g failure produces the silent-collapse and measure-zero-rescue negative results of §Discussion. No new parameter appears; each face is the same statement "$E = B$" in a different coordinate. **Figures (observational view).** `scans/topdown_macro_ratios.png` — the macro-ratio plane (R_B vs R_A, two panels $\tau_g = 10$ and 30), showing the $R_B = 1$ boundary as both the basin separator (recover/collapse) and the safe-operating boundary, with the $R_A > 1$ buffer. `scans/topdown_ratio_separation.png` — the ψ-regime closed form $R^e_A q = (1 + b_G\rho/b)/2$ (and $\psi* = 2/(1 + b_G\rho/b)$), i.e. how far the leading indicator R_A sits above the trigger R_B as the regime index $b_G\rho/b$ grows. (The τ_g cliff and full-plane no-Hopf belong to the dynamical view and are in §Numerics.) —

5 Results: principal claims

The results established by the analysis are the following; each is grounded in §4, §Numerics, and §Discussion.

- **The constant-parameter subsystem is monotonically unstable, not Hopf-driven.** Under full harvest ($\sigma = 1$), so that the equilibrium family $P = B(A)/e$ is well defined, and for the representative parameter set it has a positive real eigenvalue (≈ +0.62) for every delay, with no imaginary-axis crossing — guaranteed by the sufficient condition $S = b/b_G + G'(A*) > r$ (here $0.6083 > 0.02$), not as a parameter-free theorem; and $D(0) = 0$ on the whole equilibrium family, so there is no isolated interior attractor to frame a delay-ratio Hopf. This contrasts with the classical delayed-logistic Hopf of Hutchinson-type single- and two-delay models, in which a delay ratio produces an oscillatory onset; here the onset is a structural vicious cycle. (The $78.5yr = \pi/(2r)$ coincidence is a two-loop coincidence, not the Hutchinson $\pi/(2r)$ demographic threshold.)

- **The productivity illusion is real but conditional and narrow.** A genuine B-rising-while-A-falls window exists only for a small initial deficit (≈ 5.4 yr at $E - b_0 A_0 = 0.06$, collapsing to zero at deficit ≈ 0.075; converged RK4 — see §Demonstration). It is narrow, deficit-bounded, and transient, with a computable t_{peak} and a critical deficit.

- **Technology can increase cumulative debt.** Under overshoot, yield technology raises K, P and E, and can increase cumulative debt — a Jevons-type rebound.

- **The true threshold is emergent.** There is no threshold at `A_max/2`. The real threshold is emergent: $A_c(E)$ (fixed liability) or the MSY B_{max}; `A_max/2` is only the maximal-regeneration point.

- **The full overshoot model has an equilibrium only because of η.**

- **The full system is three delayed states (A, P, D); only the $\alpha = 0$ subsystem is 2-D; K is algebraic.**

- **The operating (decline) trigger is $R_B = 1$ (footprint = total biocapacity), not $R_A = 1$.** $dA/dt < 0 \iff E > B$; $E > bA$ ($R_A = 1$) is necessary-but-not-sufficient. A system can sit on the collapse side with $R_A > 1$ (capital-dominated) or, at long τ_g, with both ratios below 1 (silent collapse, §Discussion).

- **The debt-compounding asymmetry is a theorem under multiplicative degradation.** Under $b = (b_0 + T_b)e^{-\alpha D}$, "debt compounds without bound while technology saturates" follows from the model; it is false under an additive form.

Scenario B/C is the inverse of the orchard framing. In Scenarios B and C the environmental stock A rebounds to $\approx A_{max}$ (≈ 1.19) while population P and the harvest/biocapacity B collapse — an "environment recovers, humans collapse" outcome that is the opposite of the orchard framing ("humans die, orchard survives"). This is a scenario outcome, not a claim to be over-claimed. —

6 Falsifiable predictions

These are the testable content of the model. They fall into two registers: predictions 1–3, 6 and 7 are dynamical (stability, onset, recovery dynamics); prediction 5 straddles the two views (the ψ-mediated mask, §4.5); and prediction 8 is the observational one (the ratio safe-operating space and whether a monitor can forecast collapse, §4.5). The mapping of each prediction to the section that establishes it is tabulated in §S6.

1. **Which lag destabilises** is set by the sign of Λ/χ: on the constant-parameter subsystem (under full harvest, $\sigma = 1$) the interior point is monotonically unstable for every delay — provided $b/b_G + G'(A*) > r$, the representative-case condition — giving a positive real eigenvalue and no imaginary-axis crossing, so the "which-lag" question has no Hopf answer there.

2. The **oscillation period near onset** $\approx$ **4**$\times$ **the dominant lag** — applies only on a Hopf boundary; the constant-parameter subsystem has no oscillatory onset (monotone collapse), though the full model (with η, D) can still show damped oscillatory transients.

3. The productivity illusion has a **computable peak-biocapacity time** t_{peak} with the signature "B rising while A falls".

4. **Reducing a policy lag** τ_e **matters comparably to reducing the overshoot** f — a testable, policy-relevant ranking.

5. **The flow-share** ψ **locates the masking illusion** (§4.1, §4.5): the "B rises while A falls" window should be more visible in flow-dominated (high-ψ, $b_G\rho < b$) systems than in capital-dominated (low-ψ, $b_G\rho > b$) ones.

6. **The regeneration lag** τ_g **sets recovery vs collapse** (§Numerics): the recover fraction collapses from $\approx$ 40 % to $\approx$ 5 % across a steep **18–20 yr band**, so the model predicts collapse whenever the regeneration lag exceeds $\approx$ 20 yr and recovery only for $\tau_g \lesssim 18$. The empirically anchored effective timescale from the published recovery statistics is $\approx$ 25–33 yr (representative $\approx$ 29 yr; SI §S5.1) — at or above the collapse threshold — with primary forests at the top of that range and many soils and (non-recovered) fisheries in or beyond it. This is the necessary-but-not-sufficient reading of §4.5: $R_B = 1$ is necessary but not sufficient once τ_g is long. It is a quantitative, direct prediction tied to the empirical τ_g band.

7. **Regeneration *rate* vs *lag* are two decoupled levers** (§Discussion): time-to-50 % scales $\approx$ inversely with the regeneration rate ρ, but whether the stock recovers at all is set by the regeneration lag τ_g. A system with a long τ_g cannot be rescued by raising ρ alone.

8. **The operating boundary is** $R_B = 1$ **(footprint = total biocapacity), not** $R_A = 1$ **(§4.5).** R_A crosses 1 earlier than R_B (by $1/\psi$) but is not the collapse trigger, and its lead is $\approx$ 0 in flow-dominated systems. A system can be on the collapse side while $R_A > 1$ (capital-dominated) — and, in the long-τ_g regime, even while $R_B < 1$ (silent collapse, §Discussion). Falsifiable: monitor the two ratios and confirm the trigger is $R_B = 1$, not the earlier-crossing R_A.

The two masks. *Technology mask:* B rises while A falls (requires progress). *Liquidation mask:* E steady while A falls (requires none). The discriminating observable is exactly which holds. **Why the illusion is necessarily transient.** From $B = bA$ (flow component) the growth rate of biocapacity is $dlnB/dt = dlnb/dt + dlnA/dt$. Under the mask window A falls, so the second term is negative; the first term is bounded because technology saturates (T_b is a logistic wave, so $dlnb/dt \leq 0$ eventually). The sign of $dlnB/dt$ is therefore pinned by the unboundedly negative $dlnA/dt$ under the stock-liquidation cycle, so a B-rising-while-A-falls window cannot persist: the yield gain from a bounded technology wave is finite, whereas the stock decline it is meant to offset is unbounded once liquidation takes over. This is the analytic reason the mask is narrow and deficit-bounded (§Demonstration), and why the "productivity illusion reads as if sustainability were improving" is a transient, recover-to-reckoning signal rather than a new equilibrium. —

7 Presentation

- **Thesis (stated once here and in §4):** the balance point is $R_B = 1$ (footprint = total biocapacity); the flow-yield ratio $R_A = 1$ is a leading but non-causal signal; and when the regeneration lag is too long neither ratio warns — the lag, not the ratio, is the controlling variable.
- **Feedback diagram with a switch** (fruit harvest vs. capital liquidation): $A \rightarrow {}^{b}B \rightarrow K \rightarrow P \rightarrow E$, with $E - bA$ deciding the switch, $D \rightarrow {}^{\alpha}b$, and an exogenous bounded T_b. Two positive-feedback loops compound: Loop 1 (stock-liquidation: $A\downarrow \ \rightarrow B\downarrow \ \rightarrow deficit\uparrow \ \rightarrow liquidation\uparrow \ \rightarrow A\downarrow$) and Loop 2 (debt-erosion: $D\uparrow \ \rightarrow b\downarrow \ \rightarrow B\downarrow \ \rightarrow deficit\uparrow \ \rightarrow D\uparrow$). Rendered: `scans/feedback_diagram.png`.
- **Mappings.** In the orchard framing, the flowing fruit corresponds to biocapacity, and the standing trees correspond to the capital stock that generates it; the model's flow share ψ is the share of sustainable biocapacity that comes from the flow.
- **Non-smoothness and one-sided stability.** At the switch $E = bA$ the exact $[\cdot]_+$ is non-smooth, so the sustainable point is a boundary of the deficit regime, not an interior point. Report one-sided stability (linearise from the deficit side only). Because the exact switch yields no imaginary-axis crossing (§Numerics), the non-smoothness does not create spurious oscillations.
- **Parameter justification.** The representative values are illustrative and must be caveated: ρ is large in the original scale, and γ, α, τ are lumped/effective parameters, not independently measured.
- **Limitations of National Footprint Accounts.** The accounts are account-based (measured yields and conversion factors) and do not capture soil erosion, deforestation, or groundwater depletion, so estimated biocapacity likely overstates and the Footprint likely understates real overshoot (§Model scope).
- **Units.** Symbols and units are given for A_{max}, b_0, Δb, and every other symbol in the symbol table (§2.1); no dimensionless parameter is left without its stated unit.

Related literature. The delayed-logistic *stability* result is Hutchinson (1948), the classical single-delay logistic with its $\pi/(2r)$ demographic threshold. Haberl & Aubauer (1992) *apply* this delayed-logistic framework to human population/load dynamics; their contribution is the application, not the introduction of the time delay into population dynamics, which is Hutchinson. Brander & Taylor (1998) is a distinct Ricardo–Malthus renewable-resource model. The surplus-production/MSY curve used here is Schaefer (1954), and the fold/saddle-node with a vanishing safe basin is the catastrophic-shift structure of Scheffer et al. (2001); the empirical regeneration-lag band follows Poorter et al. (2016), Poeplau et al. (2011), Hutchings & Reynolds (2004), and Neubauer et al. (2013). On the data side, the National Footprint Accounts are documented in Wackernagel & Rees (1996), Wackernagel et al. (2002), Borucke et al. (2013), and Lin et al. (2018). These accounts are not uncriticised: Blomqvist et al. (2013), Giampietro & Saltelli (2014), and van den Bergh & Grazi (2015) raise substantive methodological objections, and the Global Footprint Network-side rejoinders appear in Galli et al. (2016). These attribution statements are given so the reader is not misled about what each source establishes. —

8 Numerics, verification, and well-posedness

- **Solver.** Method-of-steps with RK4 (or $dde23/pydelay$; Shampine & Thompson 2001), the exact $[\cdot]_+$ switch as the model states it (never a smooth ramp, because a softplus ramp leaks depletion when $E < bA$), clamping $A \geq A_{ext} > 0$, $P \geq 0$, $K \geq K_{min}$, and an explicit extinction floor (so "collapse" is a model result, not a clamp). Where a smooth variant of the $[\cdot]_+$ switch is used (reference implementation only), fix its width w and report insensitivity to it. Full-(7) runs use $\eta = 0.05yr^{-1}$; the constant- parameter subsystem S0 drops D and η, so η does not enter it.

- **Protocol and convergence.** State dt, history functions, and the Δt-convergence table (e.g. scenario A $A*$: 0.8022 at $dt = 0.5$ vs. 0.8021 at $dt = 0.05$). Each reported number carries its protocol.

- **Method sensitivity and numerical checks.** The final yield $b_{final} = 0.317$ gives $D \approx 6.76$. The endpoint D_E is method-dependent (5.26 / 6.74 / 18.70 for the frozen/crashed/un-clamped endings), so it must be reported with its termination convention. The leading Hopf delays are $\tau_g* = 85.4yr$ and $\tau_p* \approx 231yr$.

- **Measured basin shrinkage.** The qualitative "basin of attraction shrinks" is quantified. The stable fraction of the (A_0, P_0) initial-condition plane for the overshoot subsystem falls from 0.506 (no delays) to 0.042 (baseline $(\tau_g, \tau_p) = (30, 25)$), and the standard IC $(1.0, 0.1)$ flips from stable to collapse. Report the stable fraction as a function of (τ_g, τ_p); complement it with the closed-form separatrix criterion $A_c(E)$ (§4.2) — the measured fraction is the empirical witness, $A_c(E)$ the analytic boundary.

- **Contrast with the gross-harvest formulation.** The deficit-driven model (Eq. 4) is structurally distinct from the standard gross-harvest (surplus-production) formulation in which total demand is removed directly from the stock. The gross-harvest subsystem has a unique interior attractor; the deficit-driven subsystem instead has a one-sided boundary at $A \to A_{max}, P \to b_0 A_{max}/e$ and no robust interior attractor, so any population overshoot into $E > bA$ triggers the vicious-cycle collapse ($A \to A_{ext}$). This is the mechanistic reason the two formulations differ (see §Comparison of formulations).

- **Basin recompute (constant-parameter subsystem).** On the documented $A_0 \times P_0$ grid (13×16 = 208; P straddling $B* = b_0 A_{max}/e \approx 1.09$), the recover fraction falls from 39.9 % (no delay) to 5.3 % (baseline $(30, 25)$); the collapse fraction rises 60.1 % → 94.7 %. The regeneration delay triggers an A_{max}-overshoot → $E > bA$ → liquidation collapse, and the recover basin vanishes abruptly as τ_g crosses ~20 yr. A time-domain illustration (`scans/r1_recovery_vs_collapse.png`) shows the same IC $(A_0, P_0) = (1.0, 0.1)$ recovering ($A \to A_{max}$, $\tau_g = 10$) versus collapsing ($A \to A_{ext}$, overshoot $A \to 1.36$, $\tau_g = 30$); a recovery-side figure (`scans/eco_recovery_insight.png`) shows the density-dependent, sigmoidal recovery trajectory and the decoupling of regeneration rate (sets recovery time) from regeneration lag (sets recovery outcome).

- **Scenario-D threshold is near-critical, not a clean result.** $(20, 20)$ gives min $A = 0.631$ and recovers, while $(30, 25)$ gives min $A < 0.6$ and collapses ($0.6 = A_{max}/2$, the maximal-regeneration point, not a threshold). State the actual basin boundary, not a single point.

- **Reporting rigour.** (i) State the grid range of the stability scan and normalise "barely positive" $Re\lambda$ by r (a margin of $< 0.01yr^{-1}$ is not small next to $r = 0.02$). A scan over $\tau \in [0, 80]yr$ cannot rule out a single-delay Hopf, because the competing demographic-delay Hopf appears only at large $\tau_P \approx 225yr$ (slow $\omega \approx 0.011yr^{-1}$) once parameters are perturbed off the knife-edge. State the location of any "no single delay destabilises" claim as a function of Λ/χ, and ensure any such scan extends to $\tau_P \gtrsim 250yr$ before concluding. (ii) State whether interpolation occurs for the τ_g/τ_p (both exact multiples of Δt), or use a non-multiple step. (iii) Complete the scenario/parameter table (which e, which lags, whether T_b is active). (iv) Analyse the trivial equilibrium $(A, P) = (0, 0)$ and the no-recovery region. (v) Reconcile the "max Ω not reported" footnote (Ω peaks at $\approx$ 5.0 on D, $\approx$ 4.1 on E, clipped at 8.5).

- **Characteristic equation, crossing curves, and full spectrum.** We apply the exact crossing-curve formalism (Hale & Huang 1993; Gu, Niculescu & Chen 2005) and a full-spectrum computation to the constant-parameter subsystem. Linearising about an interior point $A*$ of the equilibrium family $P = B(A)/e$ (defined for full harvest, $\sigma = 1$) gives $D(s; \tau_g, \tau_p) = (s - a_1 e^{-s\tau_g} - a_3)(s + r) - a_E a_4 e^{-s\tau_p} = 0$, which has a zero eigenvalue $D(0) = 0$ on the whole family (a neutral continuum — no isolated interior attractor) and a positive real leading eigenvalue ($Re\lambda \approx +0.62$ at $A* = 0.8$) for all delays, guaranteed by the sufficient condition $S = G'(A*) + b/b_G > r$ ($0.6083 > 0.02$, and $\geq 0.57 > r$ at $A* = 0.8$). Scanning $s = i\omega$ yields no imaginary-axis crossing, and the zero-delay $a_{11} < r$ condition is violated everywhere ($a_{11} = G'(A*) + b/b_G \geq 0.57 > r$). The χ two-gain Hopf classification (derived for the interior attractor of the gross-harvest formulation) does not transfer; the crossing-curve method should be applied to the boundary equilibrium, not an interior point (`scans/r2_char_spectrum.png`, `scans/r2_a11_vs_delay.png`).

Comparison of formulations. Two distinct formulations are frequently conflated in the surplus-production / sustainability literature, and they differ qualitatively. In the **gross-harvest** (Schaefer-type) formulation, total demand is removed directly from the stock; its constant-parameter subsystem has a unique interior attractor, can be stable for a single delay, and can exhibit a two-delay Hopf ($\chi = 1$, $\tau_g* \approx 85$, $\tau_P* \approx 225$, $s = \pi/\omega$). In the **deficit-driven** formulation of Eq. (4), only the shortfall above the flow yield liquidates stock; its constant-parameter subsystem has a one-parameter equilibrium family $P = B(A)/e$ (no isolated interior attractor), a monotone positive-real eigenvalue ($\approx$ +0.62) for every delay, no imaginary-axis crossing, and a recover/collapse basin dichotomy. The two are compared directly:

	Gross-harvest (surplus-production) formulation	Deficit-driven formulation (Eq. 4)
Equilibrium	unique interior attractor	one-parameter family $P = B(A)/e$ (no isolated attractor), under full harvest $\sigma = 1$
Linearisation	interior point; $a_{11} < r$ can hold	interior point: $a_{11} = G'(A*) + b/b_G > r$ (the sufficient condition $S > r$)
Stability	stable for a single delay	monotone positive-real eigenvalue ($\approx$ +0.62) for every delay (under $S > r$), $\sigma = 1$
Hopf / onset	two-delay Hopf: $\chi = 1$, $\tau_g* \approx 85$, $\tau_P* \approx 225$, $s = \pi/\omega$	none: $D(0) = 0$ (neutral continuum); no imaginary-axis crossing
Classification	χ two-gain Hopf (which-lag)	structural vicious cycle, no χ; crossing-curve $\rightarrow$ boundary eq.
Basin	stable fraction 0.506 $\rightarrow$ 0.042	recover/collapse dichotomy 39.9 % $\rightarrow$ 5.3 %

The "balanced-with-lags" case. In the gross-harvest formulation the balanced case ($e = r_{opt}$) with lags oscillates (the χ=1 Hopf, $\tau_g* \approx 85yr$) rather than collapsing. On the deficit-driven subsystem the balanced case still lies on the continuum and its interior points are monotonically unstable, so the "no collapse at balance but oscillation with lags" dichotomy does not apply. A full-(7) recompute is required before asserting either outcome. **Empirical grounding of τ_g and the delay-response sweep.**

- **Operational definition.** τ_g is the regeneration/recruitment-response lag — the delay between a change in the environment state A and the regeneration/recruitment response that rebuilds carrying capacity ($A(t - \tau_g)$). In the model's framing, "a tree takes $\tau_g \approx 20$–$80yr$ to bear fruit." Concretely: the time from environmental change until the regeneration/recruitment response begins to restore carrying capacity, approximated by the time to reach $\approx$ 50 % of pre-disturbance productive capacity. The $\approx$ 50 % threshold excludes the early "green shoots" transient and matches the time to meaningful biomass/stock recovery in the cited field studies (forest AGB recovered $\approx$ 122 Mg ha^{-1} within 20 yr — of order half of old-growth at many sites — median $\approx$ 66 yr to 90 %; soil SOC at a new equilibrium after ~23 yr and ~17 yr). It is a conservative definitional threshold, not a fitted value; the collapse cliff sits at $\tau_g \gtrsim 20yr$ either way.

- **Field-derived band (literature banding, cited; not a full digitisation).** Forests: AGB recovery is substantial by 20 yr ($\approx$ 122 Mg ha^{-1}, i.e. of order half of old-growth at many sites), median $\approx$ 66 yr to 90 % of old-growth (Poorter et al. 2016). Soils: SOC build-up begins ~5 yr and reaches a new equilibrium after ~23 yr (deforestation) and ~17 yr (grassland$\rightarrow$ cropland) (Poeplau et al. 2011; IPCC 2006). Fisheries: rebuilding is slow and often incomplete — only ~29 % of collapsed stocks recovered to 50 % within 5–15 yr, most showed little change by 15 yr, and recovery is generally within $\approx$ 20 yr once fishing pressure is reduced to FMSY (Hutchings & Reynolds 2004; Neubauer et al. 2013). Core band $\approx$ 10–40 yr; extended $\approx$ 5–60 yr; baseline

$\tau_g = 30$ sits inside the band. Note the earlier readings "forest ≈ 50 % by 20 yr" and "fisheries ≈ 12–13 yr to 50 %" are not what these sources report; see the corrected, clearly-labelled worked conversion in SI §S5.1 ("Illustrative empirical anchoring of the regeneration timescale"), which treats the published timescales as an *effective* regeneration timescale via $\tau_g \approx 1.44 \cdot t_{50}$ (≈ 25–33 yr, representative ≈ 29 yr) and is explicitly an illustration, not a formal calibration.

- **Delay-response sweep (baseline $\tau_p = 25$).** The monotone structural instability (no Hopf, $Re\lambda \approx +0.62$) holds for every τ_g; the collapse basin is robust (≈ 5 %) across the whole field-supported band. A 1-yr-step fine sweep over $\tau_g \in [15, 25]$ resolves the transition: the recover fraction descends within a narrow band — 0.399 ($\tau_g \lesssim 17$), 0.394 (18), 0.240 (19), 0.053 ($\tau_g \gtrsim 20$) — so the "cliff" is a steep transition band ≈ 18–20 yr (half-way ≈ 0.24 at $\tau_g = 19$), not a single hard edge.

τ_g (yr)	0–17	18	19	20–60
leading Re λ	0.608–0.625	0.625	0.625	0.625 (constant)
recover fraction	0.399	0.394	0.240	0.0529

The collapse result holds for $\tau_g \gtrsim 20$ — the range the field data support for forests, soils, and many (non-recovered) fisheries — while the fast-regeneration regime ($\tau_g \lesssim 18$) returns the recovery outcome. It is not merely a one-time basin loss: §Discussion shows the recovery is not self-sustaining either. State the interval scoped, never as a flat "10–40 yr": the relevant range for the collapse mechanism is $\tau_g \gtrsim 20yr$. The field-supported band lies entirely in the collapse regime for the baseline parameter set.

- **τ_p handling and robustness.** τ_p is held at 25 yr for the sweep; the (τ_g, τ_p) grid spans $\tau_g \in 10, 15, 18, 19, 20, 30 \times \tau_p \in 10, 20, 25, 30, 40$. The cliff is governed by τ_g: the recover fraction is 0.399 at $\tau_g = 10–15$ and 0.053 at $\tau_g = 30$ for every τ_p, and at the transition the intermediate values move only slightly with τ_p (0.399–0.394 at $\tau_g = 18$; 0.245–0.269 at $\tau_g = 19$; 0.053–0.058 at $\tau_g = 20$). So τ_p shifts the band by <1 yr across a 4× range (10–40 yr) — the cliff is not a τ_p artefact. A full two-delay sweep is left to future work, but the coarse grid is no longer the only support: the flatness holds out to $\tau_p = 10$ and 40.
- **The cliff is τ_p-independent across the whole tested set, and there is no Hopf anywhere in the delay plane.** Extending the sweep to $\tau_p \in 0, 5, 10, 15, 20, 25, 40, 60$ with τ_g at 1-yr resolution: the last recovering τ_g is 18 for every τ_p (no dimensionless τ_g/τ_p tuning rescues the cliff), and a full (τ_g, τ_p)-plane scan of the characteristic roots finds zero imaginary-axis crossings with $Re\lambda \approx +0.625$ constant (`scans/topdown_delay_boundary.png`). The "basin-boundary crisis, not a fold" statement is thereby confirmed over the whole plane, not just a single-delay slice. The same computation supplies the §4.5 macro-ratio results (onset $R_B = 1.000/R_A = 1.01–1.06$; silent-collapse fraction; separator balanced accuracy; rescue set), which corroborate the neutral-continuum / no-Hopf finding from the monitoring view.
- **Conditional on baseline parameters.** The transition and recovery/collapse dichotomy are reported at the baseline ($b_G = 0.8, A_{ref} = 0.8, \rho = 0.05, A_{max} =$

$1.2, e = 0.55, r = 0.02$) and the documented IC grid (see the scenario and parameter tables in §S4). A full global sensitivity analysis over all parameters is not performed; these are conditional statements holding while the other parameters stay at baseline, and the robustness that has been checked is recorded in §S5.

- **Calibration outlook.** These are qualitative/regime claims from a literature-banded sweep, not a fitted forecast. A formal calibration (estimating τ_g, and possibly b, from a small set of well-documented case studies, or a full ML/Bayesian fit) is feasible but not required for the conceptual claims and may not sharpen the conclusions given weak delay identifiability; it is left to future work.
- **Honesty caveat.** This is a literature-banded model sweep, not a full calibration: the per-study curve digitisation is still pending, and τ_g is a lumped regeneration-response delay across heterogeneous ecosystems. The defensible claim is that the collapse mechanism operates for any τ_g in the observed band — not that τ_g equals a single measured value. Report τ_g as an interval over the observed band, not as a single number; and when a point margin is small relative to the spread of that band, do not present it as a precision result.

9 Policy extensions (Half-Earth & reservation)

- Cap the human-available flow at σB; compute both K and the additional debt from that cap; report Ω against the allocated share (not full B); state whether lags/T_b are on. This removes the impossible "$\Omega = 0.575$, $D = 0$" pair.
- Corrected cap: $K = 0.5B/e$ (not $0.5B/r_{opt}$); verified to settle at $\Omega \to 0.500$, $P = 0.217$, and a reservation policy that keeps E inside the allocated flow succeeds (no liquidation).
- **The stabilising institutional sign is the protective one.** The vicious cycle is the mobilising / extractive sign (demand responds to a perceived shortfall by raising pressure on the base); the reservation / Half-Earth cap is the protective sign (demand is held to a cap). The design principle is therefore a sign statement, not a delay statement: the stabilising institutional response is the protective / effort-reducing one, which is why the reservation (protective) policy is the candidate to pursue.
- The debt-repayment dichotomy is resolved: report the qualitative change (collapse $\to$ oscillation for $\eta \gtrsim 0.02$) and treat $\eta = 0$ as the irreversible-debt benchmark.
- **The reservation result is a nominal-path consequence.** The "E inside the allocated flow succeeds, no liquidation" result is asserted under the nominal path. To make it defensible it should be scored as a robust-viability claim: declare a disturbance class (persistent productivity/ρ/b-shocks), define a preregistered retention rule (a governance module is kept only if it improves a declared protection-and-supply score), and apply an erosion conversion (convert the declared model defect into a certified-kernel erosion margin from the closed loop's contraction rate). Until that is done, state the result as a nominal-path consequence, not a robustly-viable policy claim.

- **The reservation claim is certified only under a controller that observes the typed floor.** The result is scored against the nominal path. The reason it is only nominal is structural: the policy is certified for a controller that observes the typed stock floor A (and the allocated flow), not merely the composite B. If only an aggregate composite is observed, no observation-based policy can guarantee the stock floor — the safe controls of the states consistent with a given observation may intersect emptily, so the failure is informational (which quantitative feature of the observation design — its coarseness / aggregation relative to the typed floor — is responsible) rather than dynamical.

- **Architecture-substitution caveat.** The model analyses continuous delays τ_g, τ_p. If a policy review cadence is modelled as sample-and-hold / periodic review, do not treat it as "the delay equation sampled at T_r": the two are different operators, and moving between them can move or delete stability boundaries. Any future "review interval" extension must be a separate operator with its own monodromy, and continuous-τ conclusions must not be transferred to it.

10 Demonstration

Fig. `IMPLEMENTED_demo.png` (generated by `demo_unified.py`, the well-posed smooth-ramp solver) shows a representative overshoot run: the stock A, biocapacity B and population P all rise (a logistic overshoot / boom), then the system collapses to the extinction floor with debt D building. Two model properties are visible and reproducible: the delayed-regeneration overshoot (A briefly exceeds A_{max}, a Hutchinson-style transient here around a long τ_g/τ_p); and the debt $\rightarrow$ degradation $\rightarrow$ collapse route, in which lifting yield b raises K, P, and E, so that D accrues and b is eventually eroded. **The productivity illusion ("B rises while A falls") — quantified, converged, and conditional.**

- **Method.** The reduced masking model ($dA/dt = G(A) - ramp(E - bA)/b_G$, $dD/dt = ramp(E - B) - \eta D$, $B = bA + b_G G(A)$, $b = (b_0 + T_b)e^{-\alpha D}$, softplus ramp $w = 0.05$) was integrated with converged RK4; a 7,128-case scan plus a deficit sweep. The softplus width $w = 0.05$ is stated, and insensitivity to w is reported; a mask is scored as a contiguous span with $dB/dt > 0$ and $dA/dt < 0$, converged over $dt = 0.25 \rightarrow 0.01$.

- **Result.** A genuine mask exists, but only for a small initial deficit. For the favourable configuration ($\rho = 0.05, b_G = 0.8, b_0 = 0.5, \eta = 0.05, \alpha = 0.03, \kappa = 0.2, t_{wave} = 15, \Delta b = 1.5$, $A_0 = 1.0$), the window is 5.4 yr wide at deficit $E - b_0 A_0 = 0.06$: across it ($t \approx 1.4 \rightarrow 6.8$ yr) B rises $0.576 \rightarrow 0.647$ (peak) while A falls $0.959 \rightarrow 0.858$; over the run from $t = 0$ (where $A = 1.00$, $B = 0.578$) the peak-B rise is 0.069 and A falls by 0.142 (fig. `IMPLEMENTED_demo_masking.png`, panel a). It is converged (identical at $dt = 0.1/0.05/0.02/0.01$), so it is not an Euler artifact.

- **The mask is bounded and deficit-limited.** Window width grows with deficit up to $\approx$ 5.4 yr at deficit 0.06, then collapses to zero at deficit $\approx$ 0.075 (fig. panel b): beyond a small overshoot the stock-liquidation feedback dominates and technology cannot lift B — the run goes straight to the extinction floor with no mask.

- **Scope of the claim.** The earlier ”rising biocapacity masks a falling stock” narrative is conditional: it holds near balanced conditions, and fails under a genuine overshoot (deficit $\gtrsim$ 0.075, $\approx$ 15 % of the initial flow yield b_0A_0) — precisely the Jevons-rebound logic of §5 (technology raises K, P, E, so debt grows and closes the window). The illusion requires the wave to outpace the debt build-up — it must arrive while the stock is still high and the cumulative deficit is still small. A slow/late technology wave merely props up a falling yield.

The illusion is an instance of the compensatory-aggregation gap. The weighted composite (B, the weak-sustainability index) satisfies its aggregate floor while the typed floor — the stock A — is violated. Quantitatively it is the substitution buffer: because $B = bA + b_G G(A)$, the aggregate floor on B can be met while the typed floor on A is violated by exactly the capital growth $b_G G(A)$ — the same separation that appears in §4.5 as the gap between R_B and R_A. To say it in the model's notation: the aggregate that an index reads is a weighted sum over several floors, so a deficit in one (the stock) can be masked by a surplus in another (the yield), and a transition can satisfy the aggregate the whole way along while it never satisfies the individual floors. Two consequences follow:

- The mask is a structural failure of any aggregate that compensates a falling base with a rising yield, not an accident of one parameter set.

- Under exact-tube semantics (a transition is safe only if every state along it, not merely the endpoint, satisfies the constraints), the intermediate passage where A falls is itself a violation even if the endpoint recovers. That is a reason to report the through-time trajectory and the boundary curve rather than only an endpoint attractor.

- **Why Scenario E shows no illusion.** Biocapacity reaches $\approx$ 0.5588 in the original scenario E, but because A is rising (the logistic overshoot off the low starting population), not because of the masking mechanism. So ”no shown scenario exhibits the illusion” is the correct reading there.

- **Presentation rule.** Present the illusion as narrow, deficit-bounded, and transient (fig. `IMPLEMENTED_demo_masking.png`), with the computed t_{peak} and the critical deficit stated as numbers.

11 Model scope

This is a conceptual / stylised model with representative calibration — not a forecast. Its purpose is to make explicit the logic connecting accounting, deficit, lags, and the stock/yield decomposition, and to state which outcomes are emergent (falsifiable) versus imposed (logistic regeneration, bounded technology, the lags, the accounting identity). The 1961–2022 sentence is a qualified observation: the accounts are conservative (biocapacity likely overstated, overshoot understated); the series is consistent with the illusion reading but does not demonstrate it. **National Footprint Accounts (NFA) data limitation.** The GFN accounts are account-based — measured yields and conversion factors, no soil erosion, deforestation, groundwater depletion, or other cumulative degradation. Consequently the estimated biocapacity most likely overstates the real resource base and the Footprint most likely understates the true overshoot — the actual overshoot is likely larger than the accounts document. This is the strongest reason to treat the 1961–2022 series as "consistent with" the illusion reading rather than as evidence for it. The empirical programme the Discussion promises is the $dlnB = dlnb + dlnA$ decomposition with independent A-proxies (land cover, soil carbon, NPP). **Information-layer limit.** The signature of the illusion ("B rises while A falls") is contemporaneously observable (a read-off of two measured series). But whether that rise is a genuine recovery or a technology-driven mask is not identifiable at the observation time: identifying the mask requires knowing the technology/b-channel contribution, which is not available contemporaneously without independent proxies. Two consequences: (i) a real-time observation of B rising while A falls is not evidence for the mask mechanism over the recovery mechanism — both reproduce the same signature; and (ii) the empirical programme must target the b-channel decomposition (independent A-proxies), not the composite B series. **The empirical programme is a flux reconstruction, not a curve-fit.** The b-channel (the technology contribution to yield) is an unobserved internal flux: it is not separately measured, but it is induced by the observed stock changes. The discipline is to reconstruct unobserved internal fluxes from observed stock changes and to state which fluxes are identifiable (a flux that is not a read-out of a measured stock is not recoverable). So the programme is bounded: the b-channel is recoverable only relative to a declared observation operator, which is why independent A-proxies are needed rather than a fit to the composite B. —

12 Discussion

12.1 Structural properties and negative results

Several phenomena common in ecological models are absent here, and the absence is a structural property, not an oversight. Each is stated so the reader does not expect it:

So the collapse is robust but not predictable from a single scalar: it is driven by the regeneration lag (a finite-amplitude basin erosion), not by a fold, not by Allee low-density dynamics, and it carries no generic CSD precursor. This is more informative than a set of positive results: it tells a manager which levers do not exist (no rescue-by-stock, no early-warning signal) and which do (shorten the lag; and, only for the capital-dominated E-fold, watch for slowing). The result also inverts the classical complexity–stability expectation of May (1973), who showed that increasing species number and connectance in model ecosystems does not confer stability but tends to destabilise beyond a critical complexity threshold. Here instability is generated in the opposite sense: it emerges in a deliberately minimal, low-complexity system (two state variables, a single feedback loop, and an Allee-free regeneration term), so the collapse is not a by-product of model richness but follows from the deficit mechanism and the regeneration lag alone. Model extensions suggested by the negative results — stochasticity, seasonal forcing (to test for induced cycles), or an explicit Allee term (to test for a rescue route) — are future-work directions, not changes to the present deliberately minimal, deterministic, Allee-free model. **The macro-ratio monitor fails silently in the long-τ_g regime.** The R_B/R_A ratios of §4.5 are necessary-but-not-sufficient once the lag is too long: on the documented $A_0 \times P_0$ grid at $\tau_g = 30$, 36.5 % (72 of 197) of collapses begin with $R_B < 1$ and $R_A < 1$ — neither ratio warns; only shortening the regeneration lag helps. (At $\tau_g = 10$, only 1.6 % of collapses are silent.) Ratio-based monitoring therefore cannot replace the lag: it is

Phenomenon looked for	Present? (verified)	Why absent / what it is	Implication
Critical slowing down (CSD) / early warning before the τ_g cliff	No	the τ_g transition is a first-order basin-boundary crisis: $Re\lambda \approx +0.62$ is constant (no eigenvalue crossing zero), and the P-relaxation return time is flat ≈ 150 yr for $\tau_g = 0$–18 then vanishes abruptly at $\tau_g \approx 19$–20	do not expect a smooth warning signal before a long-τ_g collapse; monitoring a single trajectory's relaxation time will not warn
CSD before the fixed-liability E-fold	Yes, but only capital-dominated ($b_G\rho > b$)	there $B'(A*) = 0$ exactly (a genuine fold, two opposite-sign-B' fixed points merging); absent at baseline where $B(A)$ is monotone	Scheffer's "loss of resilience precedes the switch" applies only to this regime-scoped fold
Hysteresis / path dependence in the threshold	No	$A_c(E)$ is single-valued; for a fixed E there is at most one deficit-region fixed point (no bistability), so the collapse–recovery threshold is the same going up as coming down	no path memory in the threshold; recovery is immediate once the liability is lowered (the asymmetry is in time, not in the threshold)
Endogenous limit cycle (sustained boom–bust)	No	the regrow-overshoot-recollapse is a single recovery-overshoot pulse (two crossings of $0.5 \cdot A_{max}$), consistent with the monotone (no-Hopf) instability	the overshoot is not a self-oscillation; sustained boom–bust would need an external mechanism
Allee-type rescue threshold (a minimum viable stock)	No	no Allee term; the collapse at $\tau_g = 30$ is domain-wide — every A_0 from 0.02 to 1.00 (including a healthy stock $A_0 = 1.0$) collapses; the recover set is a measure-zero strip ($A_0 = A_{max}$ recovers; 1.19 and 1.21 both collapse)	you cannot rescue the system by starting from a higher stock — the lag is the controlling factor; only shortening it helps

a leading/necessary check, not a collapse forecaster. This is prediction 8. **The rescue set collapses to a measure-zero strip.** The rescue set (initial conditions that recover) is a 39.9/39.9/39.4 % A-span of 0.100–1.300 (13 distinct A_0) at $\tau_g = 0/10/18$, then collapses to 5.3 % with a 1.200–1.200 (1 distinct A_0) strip at $\tau_g \geq 20$ — i.e. at A_{max} only. Combined with the absorbing collapse (§12.2), this is the quantitative form of the "cannot rescue by starting higher" statement. **The transition is a nonlocal, not a map, bifurcation.** The asymptotic one-step map (when the delays are restored to their large-time limits) has fixed points exactly on the equilibrium family ($E = 0.15 \to A = 0.283$; $0.30 \to 0.576$; $0.50 \to 0.986$; boundary $A_{max} = 1.2$) and its local $Re\lambda_{max}$ is $+0.5917 \to +0.6249 \to +0.6250$ — constant, no flip across $\tau_g \approx 19$. The asymptotic map therefore reproduces the fixed points but not the τ_g cliff, confirming the transition is a nonlocal basin-boundary crisis, not a map bifurcation, and that the discrete and continuous pictures do not agree as τ_g is increased. This corroborates the no-CSD reading of the delay transition, which we distinguish from the E-fold of §4.2. **Separator degradation is a class-imbalance artefact unless reported by balanced accuracy.** The neutral separator $P = B(A)/e$ (i.e. $R_B = 1$) classifies the basin at balanced 99 % (no-delay) $\to$ 81 % ($\tau_g = 30$); a linear functional shows 96 % (no-delay) $\to$ 50 % ($\tau_g = 30$, chance) while its raw accuracy is 94.7 % — equal to the majority-collapse class (197/208). Report separator/ratio accuracy by balanced accuracy, never raw accuracy on the long-τ_g basin. **Recovery overshoot is real, initial-condition-dependent, and steep.** The recovery overshoot scales steeply with the lag (a fit to the computed pulse gives $ov \approx 0.0047 \cdot (\tau_g/10)^4.84$, not a single τ_g law); it is an initial-condition-dependent transient, so report it as a scoped number rather than a universal amplitude.

12.2 Recovery dynamics

- **Recovery from the extinction floor is not self-sustaining at realistic lags.** Starting from the post-collapse floor at $\tau_g \geq 20$, the stock regrows but overshoots A_{max} and re-collapses to A_{ext}, rather than settling at the sustainable boundary: A_{peak} = 1.31 ($\tau_g = 20$), 1.45 (30), 1.60 (40), 1.88 (60), and the fraction of the run spent above $0.5 \cdot A_{max}$ falls monotonically (0.16 $\to$ 0.045 $\to$ 0.033 $\to$ 0.020) — so the collapse is effectively absorbing, not a transient excursion. This matches the field picture of slow, incomplete, often-failed recovery (Hutchings & Reynolds 2004; Neubauer et al. 2013). The collapse is not merely hard to reverse, it is self-sustaining under the regeneration lag. (This is separate from the masking window and from the Allee-free fixed-liability threshold — the same delay-driven liquidation, read as a recovery-blocking mechanism.)
- **The regeneration rate sets how long recovery takes; the regeneration lag sets whether it happens at all.** Sweeping ρ and τ_g independently from the extinction floor, the time to 50 % of A_{max} scales roughly inversely with ρ (t50 $\approx$ 145/ρ yr: 203 at $\rho = 0.03$, 142 at 0.05, 105 at 0.08, 82 at 0.12, at $\tau_g = 30$), while whether it recovers at all is decided by τ_g (recovered at $\tau_g \leq 18$, collapsed at $\tau_g \geq 20$, for every ρ):

A manager has two genuinely different levers: accelerate the regeneration rate (restore stock-by-stock productivity) to shorten the time to recovery, versus shorten the regeneration lag (recruit/protect the seed source) to change whether

ρ (yr⁻¹)	$\tau_g = 10$ (recovers)	$\tau_g = 30$ (collapses)
0.03	t50 $\approx$ 162 yr	t50 $\approx$ 203 yr
0.05	106	142
0.08	74	105
0.12	56	82

recovery happens. The two do not substitute: raising ρ makes a doomed trajectory recover faster to 50 % but does not keep it from collapsing under a large τ_g.

- **Recovery overshoots the old A_{max} before settling (a catch-up rebound pulse).** From the floor, a system that does recover rises past A_{max} before relaxing: A_{peak} = 1.21 ($\tau_g = 10$), 1.25 (15), 1.28 (18) — up to +0.08 beyond A_{max} (fig. `scans/eco_recovery_insight.png`). This is the Hutchinson-style overshoot on the recovery side, and it is why a recovering system can appear to ”overshoot” briefly — a transient that must not be read as a new stable overshoot. (It is distinct from the §Demonstration collapse-regime overshoot, which ends in liquidation.)
- **Grid-dependence of the recover fraction is bounded but real.** On the same integrator, a coarse 6×8 vs fine 25×31 grid at $\tau_p = 0$:

τ_g (yr)	fine (0.05 step)	coarse (0.2 step)	fine / coarse
30	0.027	0.104	0.26
40	0.212	0.208	1.02
50	0.301	0.417	0.72
60	0.321	0.396	0.81

The coarse grid overstates the recover fraction near collapse (at $\tau_g = 30$: 0.104 coarse vs 0.027 fine), so publish the recover fraction on the fine mesh and report the coarse value only as a bound.

- **The boundary-curve result is more interpretable than a percentage.** Under no delay the separatrix in (A_0, P_0) is exactly the equilibrium family line $P_0 = B(A_0)/e$: ”recover $\iff$ $P_0 < B(A_0)/e$” matches the classification on 99.0 % of the 208-cell grid (the only exceptions are $A_0 > A_{max}$, outside the stock domain). This is the numerical witness to the neutral continuum: the recover basin is precisely the stable side of the equilibrium line. Under baseline delays $(30, 25)$ the basin collapses to a thin strip $A_0 \approx A_{max}$ (1.2) with $P_0 \lesssim B(A_{max})/e \approx 1.09$, and nothing else recovers — the delay strips away the protected below-the-line region except right at the sustainable boundary (5.3 %). Present the boundary curve rather than only the fraction.
- **The delay-response non-monotonicity at $\tau_g \approx 40$–60 is real, not a grid artifact.** The recover fraction has a deep minimum at $\tau_g = 30$ ($\approx$ 0.03) and re-opens to $\approx$ 0.21 (40), $\approx$ 0.30 (50), $\approx$ 0.32 (60); the coarse grid shows the same min$\rightarrow$ rise shape. A boundary (separatrix) curve as a function of (τ_g, τ_p) is the natural follow-up figure.

12.3 Robustness of the main results

- **Recover–collapse is robust to b_G.** On the documented grid (computed, $(30, 25)$ baseline):

b_G	recover (no delay)	recover (baseline $(30, 25)$)	Δ
0.4	0.394	0.053	0.341
0.6	0.399	0.053	0.346
0.8	0.399	0.053	0.346
1.0	0.404	0.053	0.351
1.2	0.404	0.053	0.351

Over a 3× range of the standing-stock value, the no-delay recover fraction varies only 0.394→ 0.404 (±1.3 %) and the baseline-delay recover fraction is pinned at 0.053 — the qualitative result is robust, not a b_G artefact.

- **The monotone instability is structural** (independent of ρ). The full leading eigenvalue is essentially independent of ρ (≈ +0.62, dominated by the depletion gain $a_3 = b_0/b_G$) — itself evidence that the monotone instability is structural, not a fast–slow artifact, so the stability classification of §4.3 does not carry over.

- **The fast–slow reduction is valid only for $\rho \gg r$ and we use the full equation.** The classification reduces the 2-D two-delay system to a scalar two-gain delayed logistic under the fast–slow separation $\rho \gg r$ (and while $a_{11} \leq 0$). That separation is not satisfied at realistic ρ. The table (computed at the sensitivity configuration $\tau_g = 30, \tau_p = 25, A* = 0.8, b_0 = 0.5, b_G = 0.8, e = 0.55, r = 0.02$) reports the leading real eigenvalue of the full transcendental $D(s) = 0$ alongside the fast–slow a_{11} and the timescale ratio ρ/r:

ρ (yr⁻¹)	full-$D(s) = 0$ leading $Re\lambda$	fast–slow $a_{11} = G'(A*) + b/b_G$	$a_{11} > r$?	ρ/r
0.02	+0.625	0.618	yes	1.0
0.05	+0.625	0.608	yes	2.5
0.10	+0.625	0.592	yes	5.0
0.50	+0.625	0.458	yes	25
1.50	+0.625	0.125	yes	75

Only at $\rho = 1.5$ ($\rho/r = 75$) is the fast–slow scaling justified, and even there a_{11} stays positive — no qualitative flip. At realistic $\rho \in [0.02, 0.1]$ ($\rho/r = 1$–5) the reduction is out of its regime, so the full $D(s) = 0$ must be used. We always use the full equation.

- **The masking band** is bounded by the deficit limits of §Demonstration (≈ 5.4 yr at deficit 0.06; vanishes at deficit ≈ 0.075). α and τ_p were not independently swept (each needs its own grid); state $\alpha = 0.03$ and $\tau_p = 25$ as illustration values needing calibration, and treat the ”dangerous band $(2f - 1)\nu > 1$” as an estimated indicator.

13 Limitations

- **The sustainable point is a boundary equilibrium and stability is one-sided.** The exact $[\cdot]_+$ is non-smooth at $A = A_{max}$ (and at $E = bA$), so the ”sustainable” state is a boundary of the deficit regime, not an interior point. Linearise one-sided (from the deficit/lower side only) and report that stability as such. Because the exact switch yields no imaginary-axis crossing, the non-smoothness does not create spurious oscillations. Where a smooth ramp replaces $[\cdot]_+$ (reference implementation only), fix its width w, state it, and report insensitivity to w.

- **The masking illusion is small-deficit only, not generic.** The window maxes at ≈ 5.4 yr (deficit 0.06) and vanishes at deficit ≈ 0.075 (≈ 15 % of the initial flow yield b_0A_0), and it is converged (RK4). It does not appear for deficit ≳ 0.075 and must not be cited as a general phenomenon — only as the narrow, transient, small-overshoot band of §Demonstration.

- **Parameters are representative, not estimated; the two headline results have been sensitivity-checked.** The results are illustrative and need calibration; b_G, α and the lags are lumped, not measured. τ_g is the partial exception: it is field-banded as a regeneration/recruitment-response lag (≈ time to ~50 % of pre-disturbance productive capacity), and the fine sweep shows the collapse basin and monotone no-Hopf instability are robust across that band for $\tau_g \gtrsim 20yr$ (transition band ≈ 18–20 yr), with the extended τ_p grid not moving the cliff; the fast-regeneration regime $\tau_g \lesssim 18$ returns the recovery outcome. **Fit-defect disclosure.** Any parameter pinned at an optimisation bound (a fit artefact rather than an estimate) is declared as such, never presented silently as a calibration. **Interval discipline.** A point-margin small next to an uncertainty scale is reported as a coin-flip, not as skill/stability; the same wording discipline applies to any tight margin here.

- **η is set physical (non-zero); the singular $\eta \to 0$ case concerns the full model.** The constant- parameter subsystem S0 drops D and η, so the η-singularity caveat does not apply to it. $\eta \to 0$ applies to the full (7) equation, where η decides whether an equilibrium exists. We set $\eta = 0.05yr^{-1}$, justified as a minimal environmental regeneration / degradation-removal rate. Outcomes are insensitive to η over a modest range provided $\eta > 0$; a reader setting $\eta = 0$ deliberately removes the regeneration channel and must be told it removes the equilibrium rather than perturbing it.

14 Conclusions

The model yields a sharp, policy-relevant conclusion. The balance point is the biocapacity ratio $R_B = 1$ (footprint = total biocapacity), which coincides with the neutral equilibrium family $P = B(A)/e$ and, in the capital-dominated regime, with the interior-MSY fold threshold. The commonly monitored flow-yield ratio $R_A = 1$ is only a leading, non-causal signal, and its lead over R_B is governed by the flow share ψ — so a sustainable system can legitimately run at $R_A > 1$. The more consequential result is negative. Once the regeneration lag is sufficiently long, neither ratio forecasts collapse: the transition is a nonlocal basin-boundary crisis carrying no critical-slowing-down precursor, the recover basin collapses to a measure-zero strip, and recovery from the extinction floor is not self-sustaining. The controlling variable is the regeneration lag, not the ratio; the manager's lever is to shorten the lag, and no early-warning signal or rescue-by-stock exists. Only in the capital-dominated E-fold does the classical "loss of resilience precedes the switch" structure apply, and even there it is regime-scoped. The productivity illusion — "biocapacity rises while the stock falls" — is real but narrow, deficit-bounded, and transient, and it is a structural consequence of any aggregate that compensates a falling typed floor with a rising yield. It is not identifiable contemporaneously without independent proxies for the technology channel. The paper therefore contributes a monitoring boundary ($R_B = 1$), a leading-indicator caution (R_A), and a precise statement of when monitoring fails (long regeneration lag), together with the policy-relevant levers (shorten the lag; adopt the protective institutional sign) and the levers that do not exist (no rescue-by-stock, no early warning). —

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(Data availability.) This study is a mathematical and computational analysis and reports no new empirical dataset. All numerical results, parameter sweeps, and figures were generated by the accompanying model code run from stated initial conditions and parameters; the code (see Reproducibility below) is the primary artefact, and its generated outputs are written to `data/topdown_results.json`. No primary observational data were collected. The only empirical inputs are published values used to bound the regeneration lag, which are drawn from the National Footprint Accounts and the field studies cited in the References and are not reproduced or re-distributed here; readers should consult those sources for the underlying data. **(Reproducibility.)** The model is implemented as `model_sims/corrected.py`, `model_sims/topdown.py`, `model_sims/r1_basin.py`, and `model_sims/char_eq.py` (drivers `model_sims/_run_topdown.py`, `demo_unified.py`, `mask_rk4.py`); the results are written to `data/topdown_results.json` and rendered as `scans/topdown_macro_ratios.png`, `scans/topdown_ratio_separation.png`, and `scans/topdown_delay_boundary.png`. All numeric results report their integration protocol and termination convention (§Numerics). **(Supplementary material.)** The following files accompany this manuscript and are cited where their content is used:

- `supplementary/SUPPLEMENTARY_information.md` — **Supplementary Information (S1):** the full model specification, the symbol table and units (§2.1), the non-dimensionalisation and scaling, the analytic derivations (MSY, the fixed-liability threshold, the equilibrium family $P = B(A)/e$, the stability classification), the scenario and parameter tables, the sensitivity/robustness record, the prediction$\rightarrow$ section map, and the supplementary references (§S1–§S7).
- `supplementary/FIGURE_CAPTIONS.md` — **Captions for supporting figures S1–S14**, each linked to the script that generates it.
- `supplementary/REPRODUCTION_GUIDE.md` — **Code map and reproduction guide:** dependencies, file/folder structure, and step-by-step commands to rebuild every numeric result and figure.
- `supplementary/DATA_AVAILABILITY.md` — **Data and code availability statement**, including a deposit/repository recommendation.
- `supplementary/ABSTRACT_submission.tex` — **The abstract as LaTeX** (mathematics kept as LaTeX for the submitted manuscript; 300 words).
- `supplementary/FIGURES/S1–S14.png` — **Supporting figures S1–S14** (see the captions file): S1 feedback/causal-loop diagram; S2 macro-ratio plane (R_B vs R_A); S3 flow-share separation; S4 delay-boundary / τ_g cliff; S5 representative overshoot run; S6 productivity-illusion / masking window; S7 recovery vs collapse; S8 recovery insight (rate vs lag); S9 characteristic spectrum; S10 a_{11} vs delay; S11 sustainable-yield regimes; S12 basin heatmap; S13 recover-fraction vs τ_g; S14 graphical abstract.

Declarations

Data availability

The code and data that support the findings of this study are available in the accompanying supplementary package (Supplementary Information, supporting figures S1–S14, a reproduction guide, and a data/code availability statement). A version of this manuscript together with the supplementary package is deposited at https://zenodo.org/records/22554480. This study is a mathematical and computational analysis and reports no new empirical dataset. All numerical results, parameter sweeps, and figures were generated by the accompanying model code run from stated initial conditions and parameters; the code is the primary artefact, and its generated outputs are written to `data/topdown_results.json`. No primary observational data were collected. The only empirical inputs are published values used to bound the regeneration lag, which are drawn from the National Footprint Accounts and the field studies cited in the References.

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Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author used AI-assisted tools for editorial support, including language refinement, formatting, and computational-reproducibility checks on the accompanying model code. After using these tools, the author reviewed and validated the content as needed and takes full responsibility for the content of the publication.